

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461216.719 +/- 0.073 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.098 +/- 0.039
Transit depth (Rp/Rs)^2	0.97 +/- 0.76 [%]
Orbital Inclination (inc)	81.02 +/- 2.81
Airmass coefficient 1 (a1)	1.457 +/- 0.026
Airmass coefficient 2 (a2)	-0.278 +/- 0.03
Scatter in the residuals of the lightcurve fit is	1.77 %
Best Comparison Star	None
Optimal Aperture	10.16
Optimal Annulus	21.17
Transit Duration (day)	0.029 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.098 ± 0.039 vs 0.119 (deviation 0.51σ)
Duration fit vs published	0.029 d vs 0.0942 d (deviation 2.07σ)
Mid-time O-C	6.8 min
Depth SNR	2.4
Residual scatter	1.77 %

Light curve

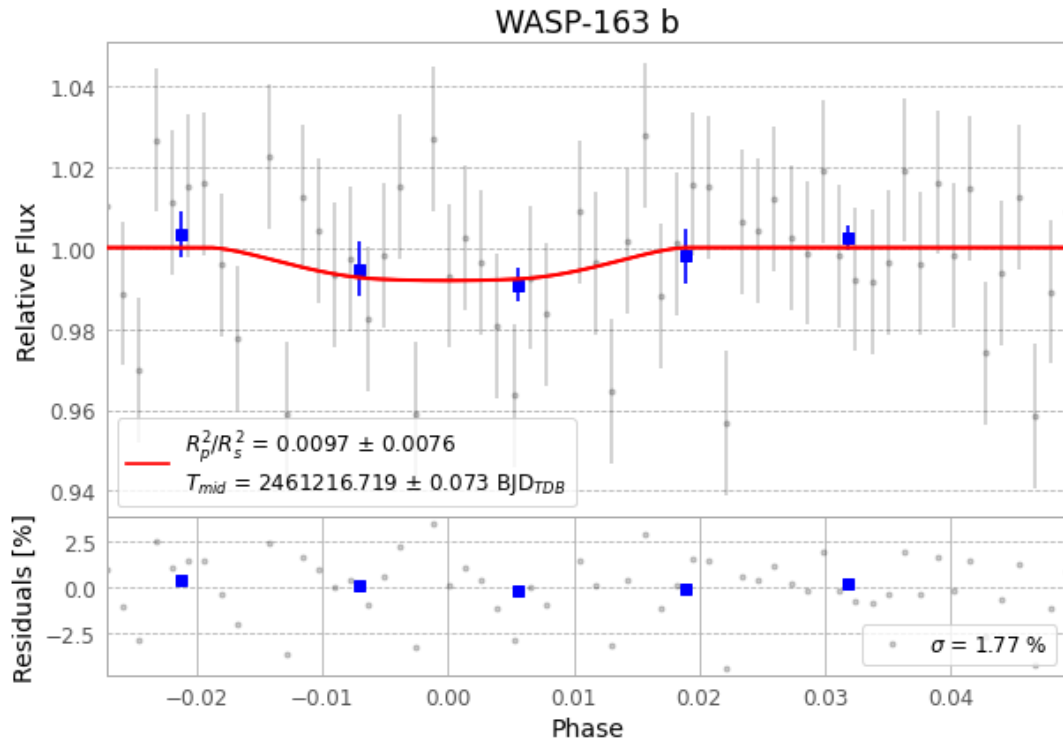


Figure 1 — FinalLightCurve_WASP-163 b_2026-06-24.png (SHA-256 9eb653085444080e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [565, 308], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 24f0a1eb83c468f3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

60 FITS frames (39 MB), WASP-163260625040316.FITS → WASP-163260625070015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-260625.log (a934389ba7a9...), FinalLightCurve_WASP-163 b_2026-06-24.png (9eb653085444...), FinalLightCurve_WASP-163 b_2026-06-24.csv (e15147b8d8f4...)

Compute resources

Wall time 415.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:20Z.